

New and Emerging Areas in Medicine: Competencies for an AI-Enabled Practice

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Disclosures

- Some of the work shared in this presentation is supported with funding from the Josiah Macy Jr Foundation
- No other relevant financial disclosures
- Resources shared are on behalf of diverse institutions, settings, and many talented individuals
- AI tools supported the development of selected session materials. The presenter directed the work, critically reviewed and revised all content, and retained full responsibility for the final materials.

Learning Objectives

- Describe the goals and structure of the AAMC AI competencies within the new and emerging areas series
- Examine implications of AI Competencies for certification, assessment and ongoing professional development across specialties.
- Identify strategies for specialty boards to incorporate AI related knowledge, skills and behaviors into standards and assessment approaches.

AAMC Mission

The AAMC leads and serves the academic medicine community to improve the health of people everywhere.

AAMC Vision

A healthier future through learning, discovery, health care, and community collaborations.





The AAMC is a not-for-profit 501(c)(3) educational association whose purpose is to advance medical education



Illinois
corporation
founded in
1876

Employment

735



Full-time employees
Competitive benefits plans
including 403(b) retirement plan

Members



Medical Schools
163 in US,
14 in Canada



Academic
Health
Systems & VAs
~500



Academic
Societies
70+

210,000 full-time faculty members, **99,000** medical students, **162,000** resident physicians, and **60,000** graduate students and postdoctoral researchers

Major Services

MCAT

AMCAS

PREVIEW

VSLO

ERAS

Journals



MedEdPORTAL
THE AAMC JOURNAL OF TEACHING AND LEARNING RESOURCES

Which role best describes your work in health care and medical education?

1. ABMS Member Board leader or staff
2. Practicing physician or ABMS-certified diplomate
3. Medical education or residency/fellowship director
4. Health system, credentialing, or medical staff professional
5. Specialty society or other professional association representative
6. Patient, public, or community representative
7. Trainee (medical student, fellow or resident)
8. Other

How would you describe your personal use of AI (large language model such as Copilot or ChatGPT) in your professional work?

1. I have never used
2. I use it occasionally for specific tasks
3. I use it extensively

How would you describe your area of focus?

1. Undergraduate medical education or UME
2. Graduate medical education or GME
3. Continuing medical education and professional development or CME/CPD



Changing Environments

Rapid advances in knowledge, society, innovations, population growth, climate change, technology, etc.

What should a physician know, do, and value in an AI-enabled health system?



Artificial Intelligence or AI

A broad set of technologies that can perform complex tasks that traditionally require human cognitive work (e.g., recognizing patterns, generating language, making predictions, and synthesizing information).

AI Applications in Health Care

- I. **Clinical Decision Support:** Augments clinical judgment with personalized, evidence-based recommendations.
- II. **Operational Analytics:** Improves system-level performance, efficiency, and resource use.
- III. **Workflow Enhancement:** Automates administrative work to maximize clinician focus on patient care.

“This year’s **#1 concern—Navigating the AI Diagnostic Dilemma**—underscores how unchecked dependence on AI tools can increase diagnostic errors, perpetuate bias, and erode critical thinking skills. Although AI has immense potential to improve clinical workflows and expand access to expertise, the rapidly growing use of AI in healthcare raises serious safety and governance challenges.”(*ECRI)

AI Applications in Medical Education

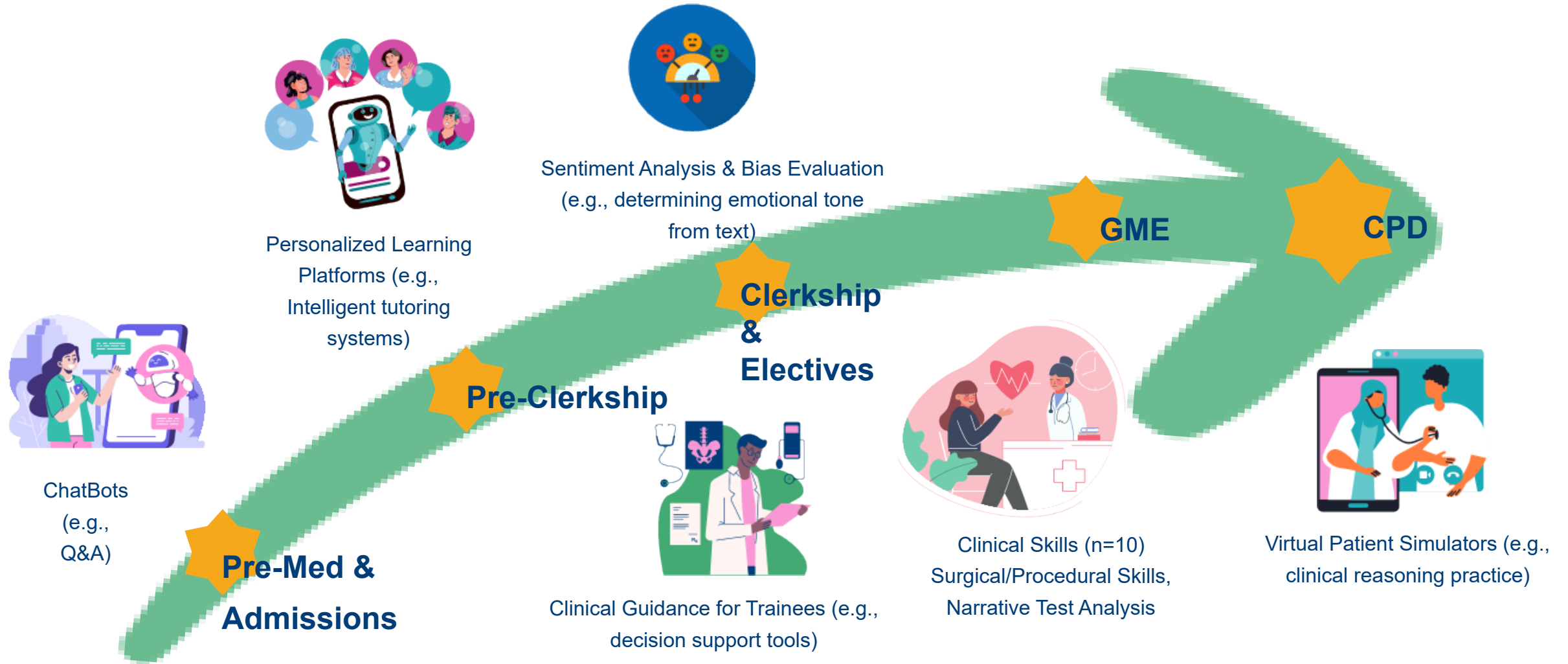
AI FOR **MEDICAL
EDUCATION**

Incorporating AI into the work you do

Teaching AI best practices to learners,
faculty, and staff

AI IN **MEDICAL
EDUCATION**

AI Applications in Medical Education

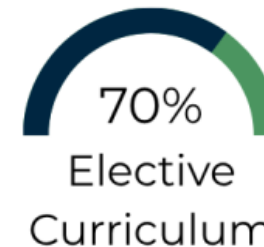
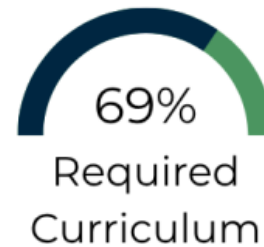


AI in U.S. & Canadian Medical School Curricula

2022-2023



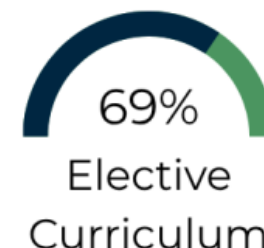
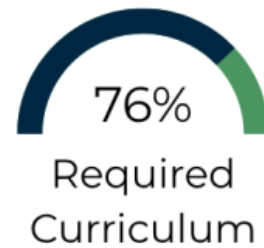
53%



2023-2024

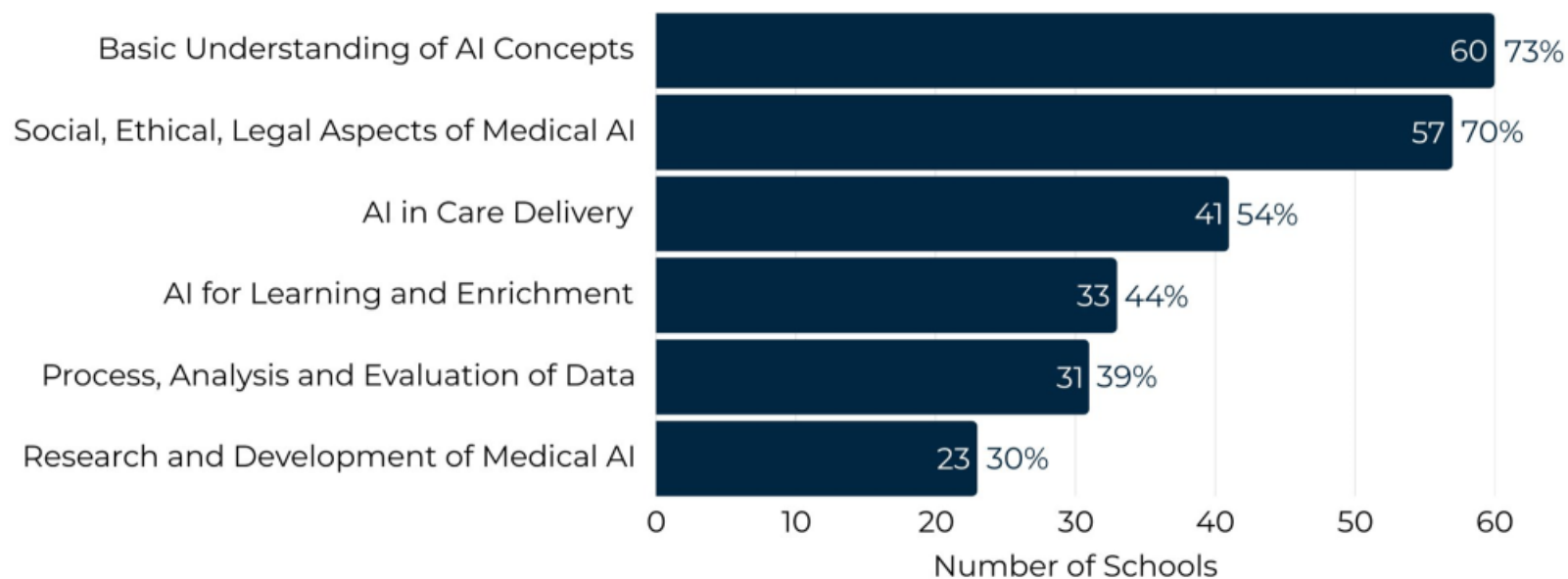


77%



Data presented from the AAMC AACOM Curriculum SCOPE Survey

A focus on the AI basics —for now



Most schools are focused on the AI fundamentals and its ethical use, but AI also is being infused in care delivery and research and development.

Data presented from the AAMC AACOM Curriculum SCOPE Survey

AAMC's competencies for New and Emerging Areas in Medicine Series



2019 Quality Improvement & Patient Safety
(Version 2.0 Published 2024)

2021 Telehealth ***Version 2.0 Just Released!**

2022 Diversity, Equity and Inclusion

Artificial Intelligence **Coming Soon!**



New and Emerging Areas in Medicine Series



As we develop AI competencies for physicians...

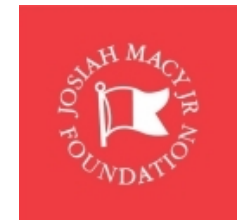
What do we expect a:

- **Novice**
- **Proficient**
- **Expert**

...to *know, do, and value* in an AI-enabled health system?

Overall Project Purpose

- **To develop a national set of Artificial Intelligence (AI) competencies for medical education to support learners along the developmental continuum.**
- This work will help ensure all medical learners – regardless of where they train – gain foundational knowledge in AI principles, understand its ethical and legal implications, and are equipped to collaborate with data scientists and technologists in healthcare settings. Ultimately, this will enable physicians to deliver safe, effective care, while also leading in an evolving digital healthcare landscape.



Artificial Intelligence Competencies: Working Group



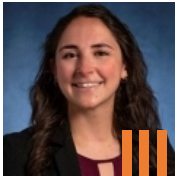
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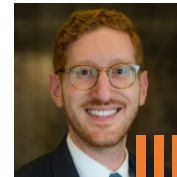
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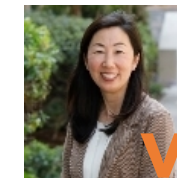
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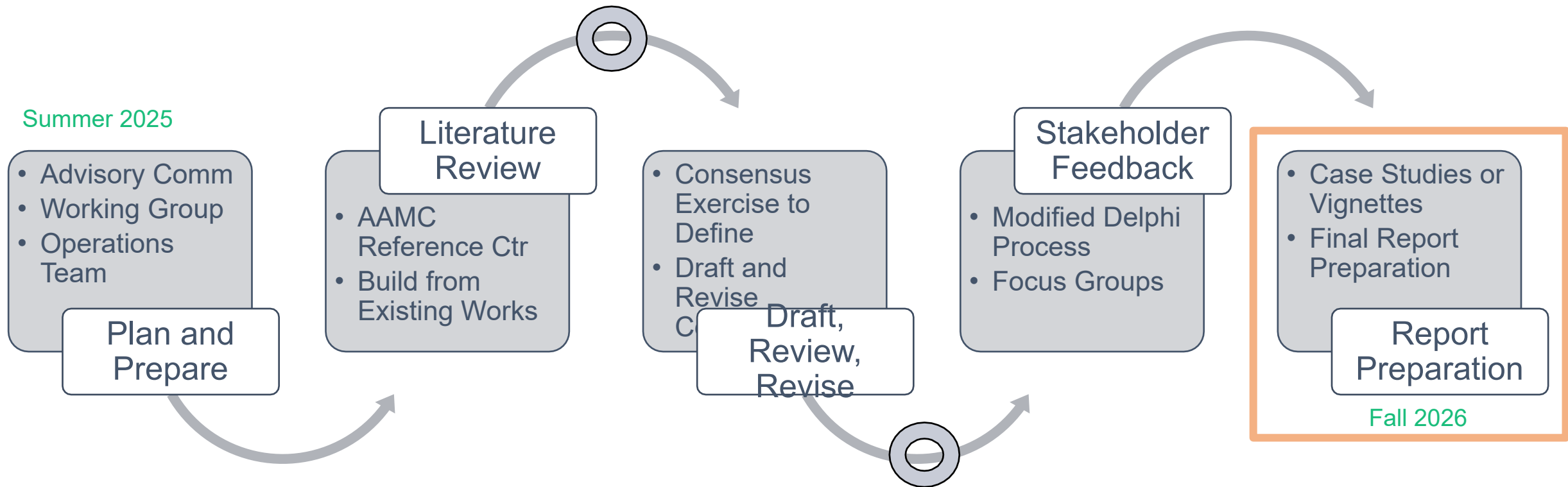


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Artificial Intelligence Competencies Across the Learning Continuum



The activities are supported (in part) by the Josiah Macy Jr. Foundation.

NOTE: AI tools will be used to support development (e.g., literature review, generating, summarizing sources, analyzing public comments and survey data, and more).

Artificial Intelligence Competencies: Community Co-Creation

Feedback on the draft competencies occurred through two rounds of structured online surveys involving early reactors and public reviewers as well as focus groups (virtual and in person).

Online Surveys

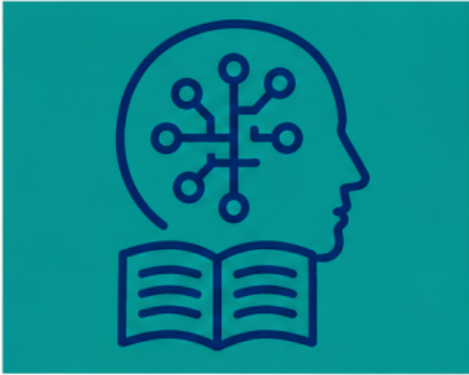
Draft 1 = 683

Draft 2 = 308

Focus Groups

- + Four regional meetings of the AAMC Group on Educational Affairs (NEGEA, SGEA, CGEA, WGEA)
 - + AAMC Together We Lead Conference
- + AAMC/AMEE/NUS/IAMSE International Advisory Committee for AI (IACAI)
 - + AAMC/Manatt Digital Health Collaborative

Artificial Intelligence Competencies Across the Learning Continuum



Knowledge



AI Assisted Care and
Clinical Reasoning



Communicating
and Collaborating



Professionalism
and Ethics



Practice Based
AI Learning and Improvement

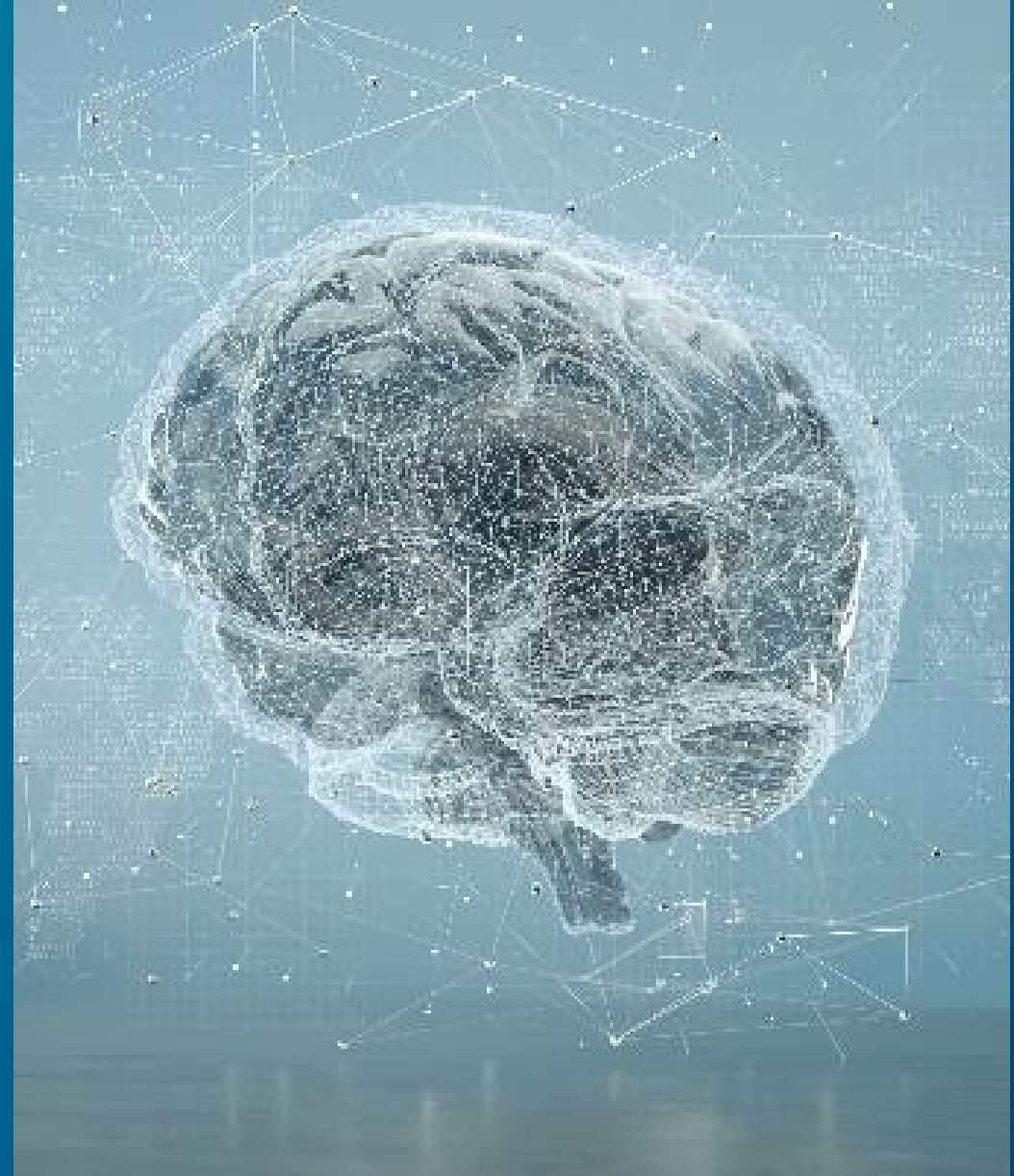


Systems Based
AI Practice

What should a physician know, do, and value in an AI-enabled health system?

I: AI Knowledge for Practice (n=3)

Explain what artificial intelligence is, how it works, and how its evolving tools and applications can be critically and responsibly used in health care, education, and research.



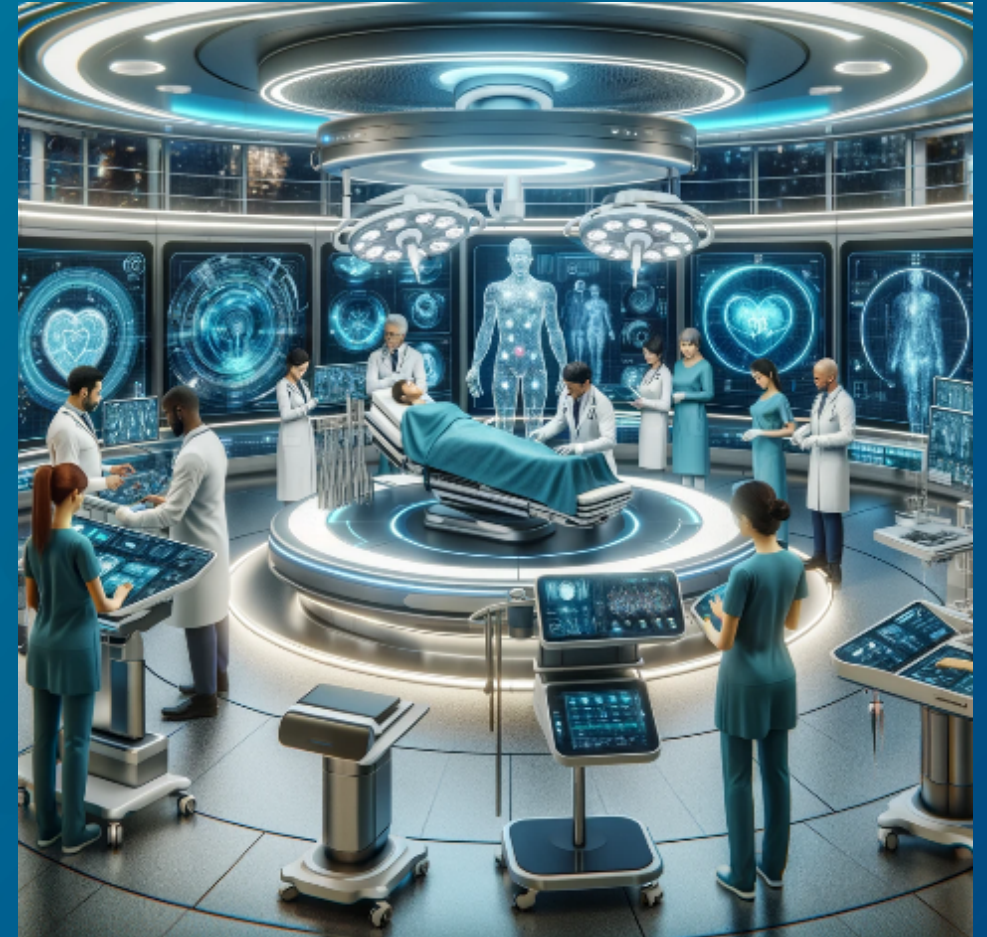


II: AI-Assisted Patient Care & Clinical Reasoning (n=5)

Use AI with clinical discernment to assist and augment clinical reasoning, patient education, and care delivery while ensuring safe, equitable, patient-centered care.

III: Communicating about AI & Interprofessional Collaboration (n=5)

Communicate about AI-enabled processes, outputs, and limitations with patients, caregivers, colleagues, learners, and interdisciplinary collaborators to support understanding, trust and responsible use.





IV: Professionalism & Ethics (n=4)

Demonstrate ethical, legal, social, and professional awareness in the development, deployment, and use of AI, applying humanism and the principles of autonomy, beneficence, nonmaleficence and justice while preserving the professional identity, responsibilities, and trustworthiness of physicians.

V: Practice-Based AI Learning & Improvement (n=4)

Uses AI to support lifelong learning, reflective practice, critical thinking and practice improvement while maintaining expertise and humility.





VI: Systems-Based AI Practice (n=5)

Evaluates, monitors, and improves AI integration across health systems to enhance quality, safety, access, and equity through physician engagement, co-development, critical appraisal, and safeguards.

What's in a domain?: AI Competency Application Exercise

1. Define the challenge: **What contemporary AI-related issue is emerging in your specialty, and why does it matter for patient care or professional practice?**
(see examples or use your own)
2. Apply the domains: **Which AI Competency Domains are most relevant? What would competent performance look like for a physician in your specialty related to the challenge?**
3. Identify a next step: **What is one realistic action your specialty or professional community could take to advance both education and practice and address this challenge?**

Report-out statement: “For the challenge of _____, our priority competency domain/s is _____. One way to advance education or practice is _____. We would see improvement by _____”

What's in a domain?: AI Competency Application Exercise

Sample specialty-specific challenges

1. Using AI-generated clinical documentation while maintaining accuracy and accountability
2. Evaluating AI-supported diagnostic or treatment recommendations
3. Communicating with patients about the use and limitations of AI
4. Recognizing bias or inequitable performance in specialty-specific AI tools
5. Protecting patient information when using generative AI
6. Preventing overreliance on AI and erosion of clinical skills
7. Determining when AI use should be disclosed, documented, or reviewed

Worksheet: What’s in a Domain? AI Competency Application Exercise

1. Define the challenge. What contemporary AI-related issue is emerging in your specialty, and why does it matter for patient care or professional practice? (Use a sample challenge or your own.)

2. Apply the domains. Mark the domains most relevant to your challenge, then describe for that 1+ what competent performance would look like for a physician in your specialty.

Priority?	AI Competency Domain	What competent performance looks like in our specialty
	I. AI Knowledge for Practice	
	II. AI-Assisted Patient Care & Clinical Reasoning	
	III. Communicating about AI & Interprofessional Collaboration	
	IV. Professionalism & Ethics	
	V. Practice-Based AI Learning & Improvement	
	VI. Systems-Based AI Practice	

3. Identify a next step. One realistic action your specialty or professional community could take to advance education and practice_____

4. Invite 1 to serve as a reporter. Report-out statement: “For the challenge of _____, our priority competency domain/s is _____. One way to advance education or practice is _____. We would see improvement by _____.”

Artificial Intelligence

Check out what the AAMC has to offer!

- Monthly webinars and skill building workshops
- AI-enhanced tools for exploring AAMC webinars
- Dedicated virtual community
- Principles for the Responsible Use of AI in Medical Education and in Admissions and Selection
- Key resources collection
- Competencies (coming soon!)
- Key scholarship in Academic Medicine and MedEdPORTAL

aamc.org/AI



Principles for the Responsible Use of Artificial Intelligence in and for Medical Education

- 1) Maintain Human-Centered Focus
- 2) Ensure Ethical and Transparent Use
- 3) Provide Equal Access to AI
- 4) Foster Education, Training, And Continuing Professional Development
- 5) Develop Curricula through Interdisciplinary Collaboration
- 6) Monitor and Evaluate

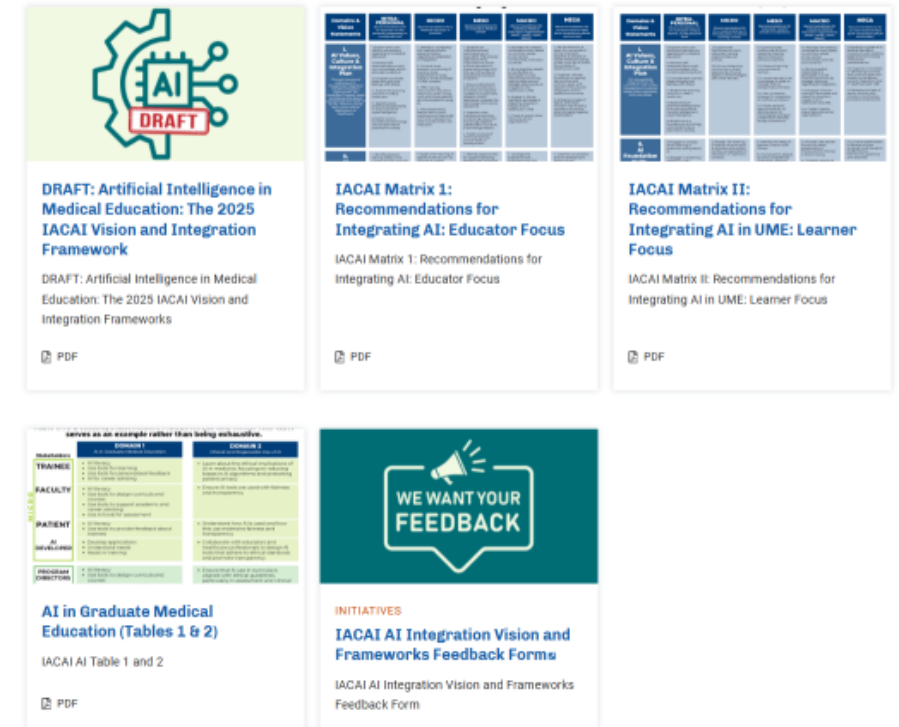


International Vision & Integration Frameworks

The International Advisory Committee for Artificial Intelligence (IACAI) created an international future vision of AI in medical education, along with accompanying frameworks, to guide proactive integration.

- Sets forth a proactive vision and call to action for responsible and ethical use of AI in medical education.
- Provides guidance across multiple domains, for various teaching and learning contexts (classroom-based education, clinical training, etc.) and targeting specific audiences (educator, learner, etc.).
- Highlights actions that individuals and institutions can take.

The IACAI is a collaboration between AAMC, AMEE, IAMSE, APMEN, and AAHCI.



medbiq.org

New Community Resource: AI Policy Development Checklist



The AAMC created this checklist to help institutions develop or update policies to address the use of artificial intelligence (AI).

This checklist was created using sample policies and resources shared by respondents to the 2024 AAMC AACOM Curriculum SCOPE Survey.



Thank you for the
opportunity to learn
with you today and
for all you do to
advance physician
development

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